Embedded Control for Robotics

Lecture # 15 State Space Representation

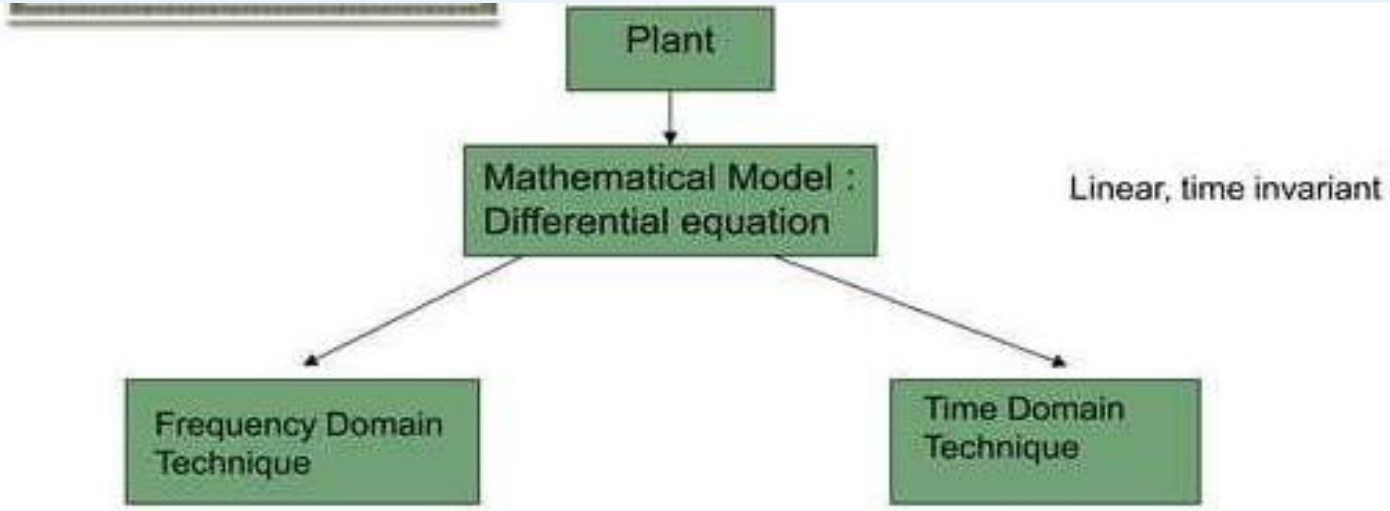
Book: Embedded Digital Control with Microcontrollers

Dr. Suleman Khan
CS, NUCES-FAST, Islamabad

Overview Today's Lecture

- State Space Representation
- Transfer Function VS State Space
- State Space Example
- Pole/Zero and Stability
- Python Demo

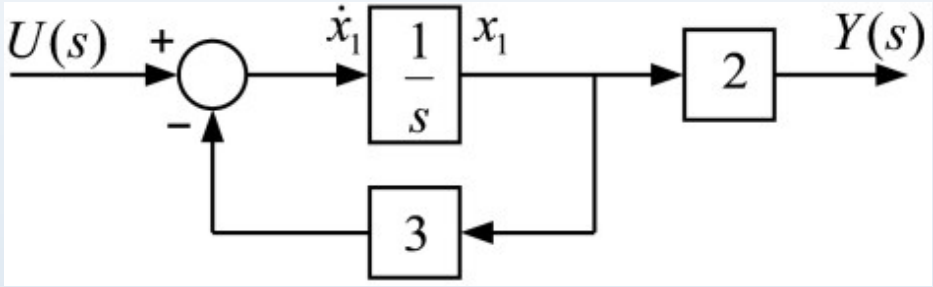
State Space Representation



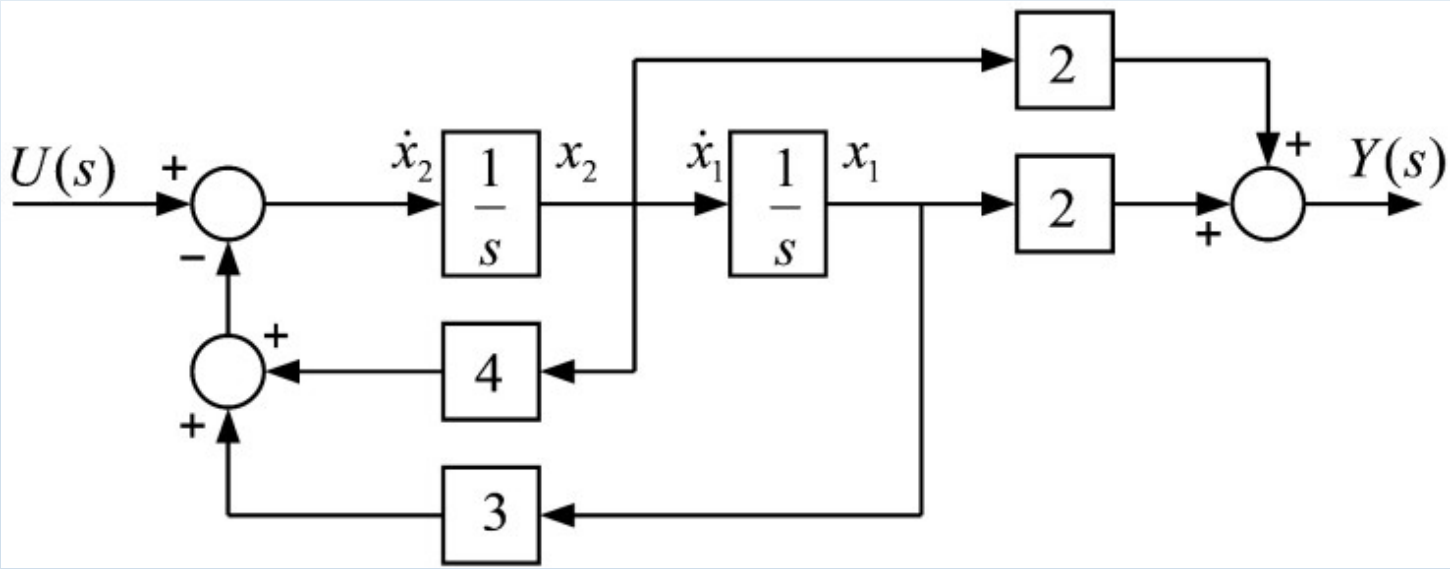
Two approaches for analysis and design of control system:

1.Classical Technique or Frequency Domain Technique.(T.F. + graphical plots like root locus, bode etc.)
2.Modern Technique or Time Domain Technique (State variable approach).

Transfer Function



$$G_1(s) = \frac{2}{s+3}$$



$$G_2(s) = \frac{2s+2}{s^2+4s+3} = \frac{2(s+1)}{(s+3)(s+1)} = \frac{2}{s+3}$$

TF VS State Space

- Both systems have same TF but very different internal representation
- TF cannot capture all detail of the internal **states of the model**
- States are the minimum set of parameters of a system which completely describe its status in time.

Why State Space Representation?

- can handle Multiple Inputs & output systems
- allow for a more geometric understanding of the dynamics of the systems,
- are numerically efficient to solve,
- handle complex systems,
- and form the basis for much of modern control theory

State Space Representation

- A state-space model represents a system by
  I. a series of first-order differential state equations
  II. algebraic output equations.

- Differential equations have been rearranged as a series of first order differential equations.

$$\dot{\vec{x}} = A\vec{x} + B\vec{u}$$

$$y = C\vec{x} + D\vec{u}$$

$\mathbf{x}$ = state vector
$\dot{\mathbf{x}}$ = derivative of the state vector with respect to time
$\mathbf{y}$ = output vector
$\mathbf{u}$ = input or control vector
$\mathbf{A}$ = system matrix
$\mathbf{B}$ = input matrix
$\mathbf{C}$ = output matrix
$\mathbf{D}$ = feedforward matrix

State Space Representation

1) Identify ***input*** variables (actuators and exogenous inputs).
2) Identify ***output*** variables (sensors and performance variables).
3) Identify ***state*** variables. (Hmmm…how ? – indep. energy storage)
4) Use first principles of physics to relate derivative of state variables to the input, state, and the output variables.

State Space Representation

- **Consider the CT differential equations:**

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = b_0 v(t)$$

- **A second-order differential equation requires two state variables:**

$x_1(t) = y(t)$ $\quad$ $x_2(t) = \dot{y}(t)$ taking derivative of x1(t) as

$\dot{x}_1(t) = \dot{y}(t)$ $\quad$ $thus$ $\quad$ $\dot{x}_1(t) = x_2(t)$

- **We can reformulate the differential equation as a set of three equations:**

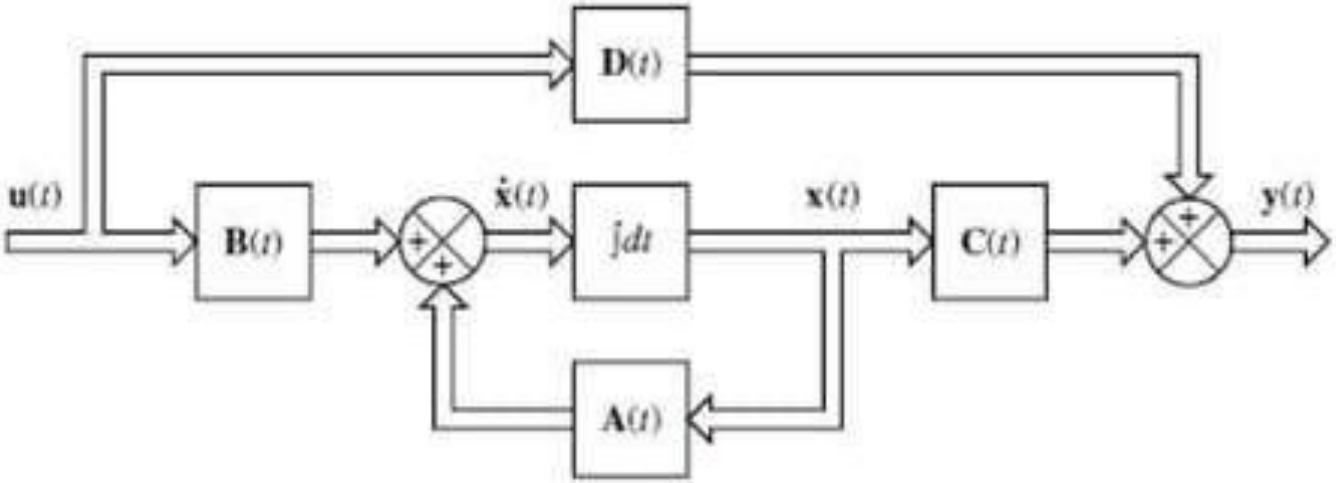
$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -a_0 x_1(t) - a_1 x_2(t) + b_0 v(t)$$
$$y(t) = x_1(t)$$

- **We can write these in matrix form as:**

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ b_0 \end{bmatrix} v(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Block Diagram of SS Eqau.



Example # 1: Diff Equn to SS

Consider the following system where $u(t)$ is the input and $\dot{x}(t)$ is the output.

$$\dddot{x} + 5\ddot{x} + 3\dot{x} + 2x = u\,, \mathrm{y} = \dot{x}$$

Can create a state-space model by pure mathematical manipulation through changing variables

$$x_1 = x,\ x_2 = \dot{x},\ x_3 = \ddot{x}$$

Resulting in the following three first order differential equations (ODEs)

$$\dot{x_1} = x_2,$$

$$\dot{x_2} = x_3,$$

$$\dot{x_3} = -5x_3 - 3x_2 - 2x_1 + u$$

Continue…

State Equations

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -5x_3 - 3x_2 - 2x_1 + u$$

Output Equation

$$y = x_2$$

System has 1 input (u), 1 output (y), and 3 state variables (x_1, x_2, x_3)

Continue…

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -5x_3 - 3x_2 - 2x_1 + u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -3 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

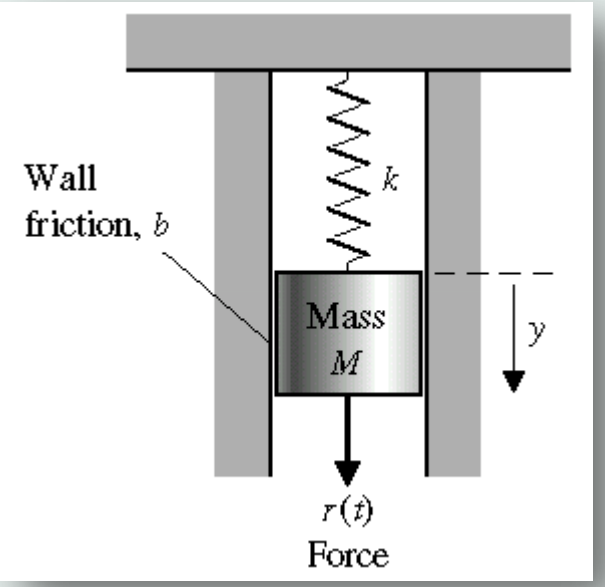
$$y = x_2$$

$$[y] = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + [0]\, u$$

Mass-Spring-damper system

Dynamic equation of the system:

$$M\frac{d^2y(t)}{dt^2}+b\frac{dy(t)}{dt}+ky(t)=r(t)$$



Therefore, we define variable x_1 and x_2.

$$x_1 = y$$
$$x_2 = \dot{y}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \frac{-b}{M}x_2 - \frac{k}{M}x_1 + \frac{1}{M}u$$

Mass-Spring-damper system

General State-Space Model:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

In matrix form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u$$

If the measured output of the system is position, then we have:

$$y = x_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

State Space Representation of RC

(a) the state variables are $v_2(t)$ and $\dot{v}_2$

$$x_1 = v_2(t)$$

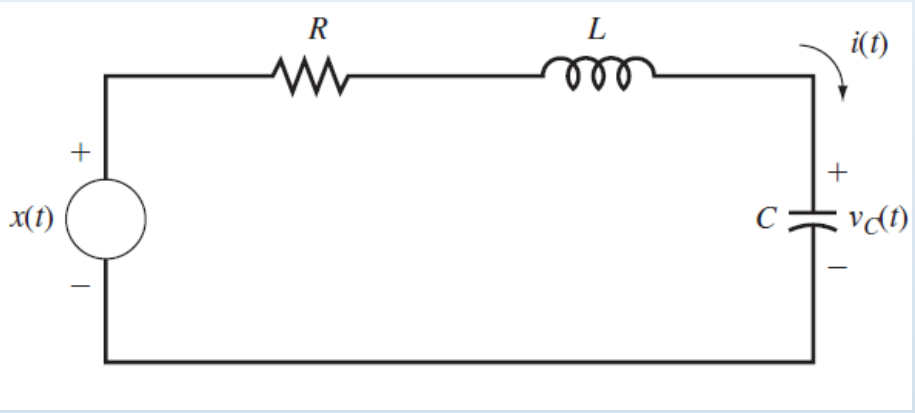
$$x_2 = \dot{v}_2 = \dot{x}_1$$

$$LC\frac{d^2v_2}{dt^2} + RC\frac{dv_2}{dt} + v_2 = v_1(t)$$

$$\dot{x}_1 = x_2$$

$$x_2 = -\frac{1}{LC}x_1 - \frac{RC}{LC}x_2 + \frac{1}{LC}u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{1}{LC} & \frac{-R}{L}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{LC}\end{bmatrix}u$$



State Space Representation of RC

(b) the state variables are $v_2(t)$ and $i(t)$.

$$L\frac{\mathrm{d}i}{\mathrm{d}t} = -v_2(t) - Ri(t) + v_1(t)$$

$$C\frac{\mathrm{d}v_2}{\mathrm{d}t} = i(t)$$

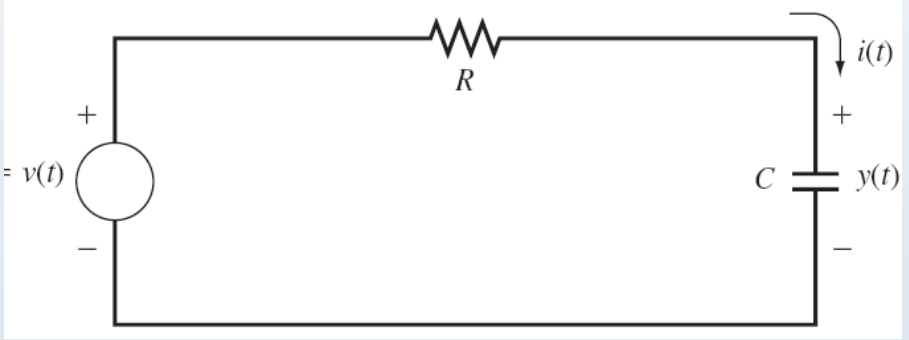
$$\dot{x}_1 = \frac{1}{C}x_2$$

$$\dot{x}_2 = -\frac{1}{L}x_1 - \frac{R}{L}x_2 + \frac{1}{L}u$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{1}{L} & -\frac{R}{L}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{L}\end{bmatrix}u$$

State Space Representation of RC

$$\frac{dy(t)}{dt}+\frac{1}{RC}y(t)=\frac{1}{RC}v(t)$$



$$\mathbf{A}=\left[-a_0\right]=\left[-\frac{1}{RC}\right]$$

$$\mathbf{B}=\left[b_0\right]=\left[\frac{1}{RC}\right]\quad \mathbf{C}=\left[1\right]\quad \mathbf{D}=0$$

$$\begin{bmatrix}\dot{x}_1(t)\end{bmatrix}=\begin{bmatrix}-\frac{1}{RC}\end{bmatrix}\begin{bmatrix}x_1(t)\end{bmatrix}+\begin{bmatrix}\frac{1}{RC}\end{bmatrix}v(t)$$

$$y(t)=\begin{bmatrix}1\end{bmatrix}\begin{bmatrix}x_1(t)\end{bmatrix}$$

Example # 2: Diff Equn to SS

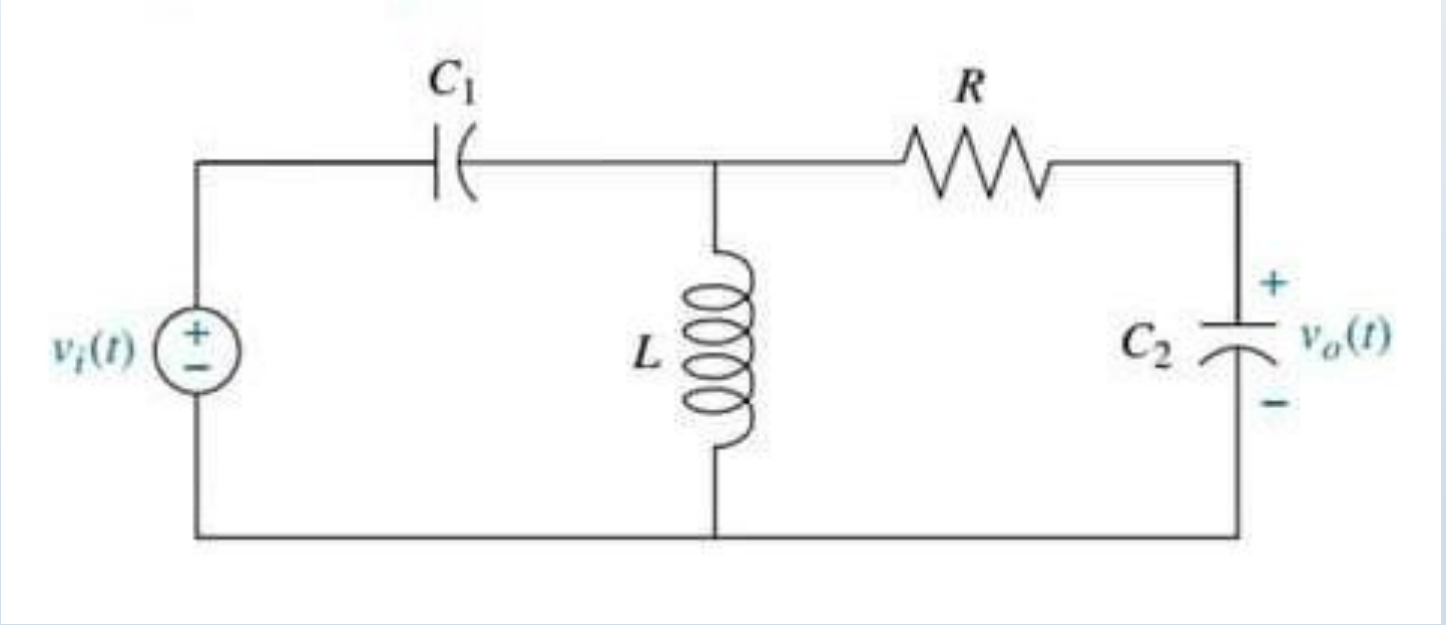
$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = u(t)$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$y(t) = x_1(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

Assignment

Find TF and SS model of the Given Circuit as a Voltage IN and out



SS to TF

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

Convert to TF

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0]\, x$$

SS: Discrete Time

$$\mathbf{x}[n+1] = \mathbf{F}\mathbf{x}[n] + \mathbf{G}u[n]$$

$$y[n] = \mathbf{C}\mathbf{x}[n] + \mathbf{D}u[n]$$

Differential Equation to SS

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = u(t)$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$y(t) = x_1(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

SS in Python

$$\mathbf{x}[n+1] = \begin{bmatrix} 1.8188 & -0.8224 \\ 1.0 & 0.0 \end{bmatrix} \mathbf{x}[n] + \begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix} u[n]$$

$$y[n] = \begin{bmatrix} 0.0360 & 0.0017 \end{bmatrix} \mathbf{x}[n] + 0.0094 u[n]$$

SS in Python


```
import mpcontrol
import mpcontrolPC

Ts = 0.0005

F = [[1.8188, -0.8224], [1.0, 0.0]]
G = [[1.0], [0.0]]
C = [[0.036, 0.0017]]
D = [[0.0094]]

ss = mpcontrolPC.ss(F, G, C, D, Ts)

N = 1000

u = mpcontrol.step_signal(N, 1)

y = mpcontrol.lsim_ss(ss, u, N, 0)
```

TF to SS in Python


```
import mpcontrolPC
import control

num = [165702.47]
den = [1, 390.2, 15701]

tfs = mpcontrolPC.tf(num, den)

Ts = 0.0005
tfz = mpcontrolPC.c2d(tfs, Ts)

tf_cl = mpcontrolPC.mptf_to_tf(tfz)
ss_cl = control.tf2ss(tf_cl)

ss = mpcontrolPC.ss_to_mpss(ss_cl)
```

SS to TF in Python


```
import control

#System Description in State-Space
F = [[1.8188, -0.8224], [1,0]]
G =  [[1], [0]]
C = [[0.0360, 0.0017]]
D = [[0.0094]]

Ts=0.0005

#Transfer Function of the System
tf_system = control.ss2tf(F,G,C,D,Ts)

print (tf_system)
```

SS Model in Python